TETHER: Tethered Causal Temporal Refinement for Frozen Surgical Stereo

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Abstract— **Modern stereo estimators are applied independently to each frame and can fluctuate under specularities, weak texture, tissue deformation, tool occlusions and camera motion. Video stereo methods fix this by building temporal reasoning into the estimator, so a new backbone means a new temporal model. We ask the opposite question: can one causal temporal module be trained once and attached, unchanged, to frozen and heterogeneous stereo estimators?**

We present TETHER, a causal plug-and-play refiner that answers it. One frozen $154{,}874$-parameter checkpoint, trained on three estimators and one dataset, improves *every* estimator on *every* arena we evaluate: five architectures, two excluded from its training, over development, held-out and out-of-domain splits — all fifteen backbone–arena cells in mean EPE, with the sequence-level exceptions named where they occur. Zero-shot on DRENDS, which no part of the module has seen, disparity EPE, Bad3 and metric depth MAE and RMSE improve in all 25 recording–backbone cells, by -5.41, -19.51, -4.02 and -3.47%, nothing adapted or tuned for the domain; the two unseen estimators we cannot separate from the three seen ones, which is the claim.

The module reads only what a black box emits: preceding disparity motion-aligned with frozen SEA-RAFT, and a shared 38-channel evidence space of disparity, motion and image geometry — no learned appearance features, no frozen backbone, no future frame — feeding a reset-and-fusion head that predicts a bounded interpolation between the current estimate and that aligned memory. One of its inputs, a forward–backward flow-consistency cue, is not read at inference, so the module deploys at $\sim$16 of its $\sim$29 ms on the unchanged checkpoint. Under a matched state machine the gain is not temporal smoothing: 3.69% mean EPE against 0.59% for the best fixed or EMA blend, which buys its mean error with bad pixels while the learned gate does not. Accumulated into a map the refined geometry reduces world-frame disagreement in all 20 held-out cells and increases it on development; we name the term consistent with which, and its limits. We additionally introduce a refinement-centric protocol scoring introduced error alongside recovered error, and apply it to our own claims.

I. Introduction

Metric three-dimensional perception underpins camera navigation, tool–tissue distance estimation and deformation analysis in robot-assisted surgery, and stereo endoscopy is attractive because depth comes from images the platform already provides. It is also hard: endoscopic scenes carry weak texture, non-Lambertian tissue, specularities, blood, smoke, deformable anatomy, rapid viewpoint change and tool occlusion [17], [18]. Modern estimators [10]–[14] give increasingly strong frame-wise disparity, but nothing in a frame-wise model requires its predictions to cohere over time. Video stereo methods build temporal reasoning into the model [1]–[4], at the price of a temporal architecture, model-specific representations, future observations, or co-design with the estimator.

We study a different deployment setting: *can one causal temporal module be trained once and attached unchanged to heterogeneous frozen stereo estimators?* Backbone and temporal logic can then evolve independently, a new estimator replacing the previous one without the module needing its cost volume, hidden state, confidence head or training pipeline. We keep one inductive bias from the video-stereo literature — decide whether motion-aligned history should influence the current estimate, and constrain the correction to a convex interpolation between the two — and redesign everything around it for a black-box interface.

Two questions follow. Does the learned head add anything simple temporal averaging would not? Matched blends recover only a small fraction of its gain while an oracle shows the aligned memory carries far more than the policy exploits, so the core problem is not obtaining temporal evidence but deciding how much of it to use. And how should such an intervention be evaluated? Smoothness is not correctness, and lower mean EPE hides whether refinement introduces new bad pixels or rare catastrophic frames, so we score relative to the frozen prediction — and, because what a robot consumes is a map, carry the same intervention into an accumulated reconstruction and measure it there.

Our contributions are three.

(i) **One module, every backbone, every domain.** We measure a single frozen checkpoint across the full backbone $\times$ domain matrix — five estimators, two excluded from its training, over development, held-out and out-of-domain splits, including the joint unseen-backbone, unseen-domain cell usually left unmeasured — and it improves all fifteen cells, with the unseen estimators we cannot separate from the seen ones.

(ii) **A black-box interface that makes that possible.** We formulate surgical temporal stereo refinement as a *causal plug-and-play adaptation problem* over frozen, heterogeneous estimators, needing neither future frames nor backbone internals, and instantiate it as TETHER: a $154{,}874$-parameter reset-and-fusion head over causal SEA-RAFT alignment, a shared 38-channel external evidence space, support-confined convex fusion and bounded corrected-state recurrence. Under a matched state machine the gain is not temporal smoothing.

(iii) **A protocol that scores the intervention, not the prediction.** We measure introduced error alongside recovered error under fixed, prediction-independent support, and it changes conclusions: on development, every spatially uniform blend recovering a meaningful share of the gain buys

it by *raising* the bad-pixel rate, while the one unlearned rule that varies in space does not. That trade is a property of the split as much as of the policy: out of domain, where the raw prediction is weak, none of the uniform blends pays in bad pixels either. Applied to our own claims the protocol is equally unkind: degenerate history-replay wins the no-reference metric, the head's weight inverts its relation to harm across splits, and what a map fuses improves on one split while degrading on the other, for a reason we can name.

II. Related Work

a) Deep stereo matching.: Stereo has moved from cost-volume regression to iterative recurrent estimation [10], [11] and broad-domain zero-shot models [12]–[14]. Our work is orthogonal: we neither modify the estimator nor assume an architecture.

b) Temporal stereo and depth.: CODD introduced the reset/fusion mechanism we build on [1]; DynamicStereo [2], PPMStereo [4] and TC-Stereo [6] follow, all co-designed with the estimator, so a new backbone means a new temporal model. Match Stereo Videos exposes BiDAStabilizer as a plug-in on a frozen estimator [3], but propagates bidirectionally. Monocular video depth is made consistent by diffusion [23] or stabiliser networks [15] — a different output space, none causal at clip level — while online weight adaptation [24] is the opposite of our constraint.

c) What can actually be compared.: We are aware of no published method that is simultaneously causal, black-box and estimator-agnostic. CODD [1], whose two-gate decision we build on, is causal but not black-box: its fusion network consumes matching features from its own stereo network, which a cached third-party backbone does not expose. The 142-channel configuration of Sec. V-J began as the closest reproduction our interface admits — its projection replaced by our warp, a frozen ResNet-18 standing in for the estimator features — so Table III is the nearest thing to running CODD here. Among black-box candidates StereoDiffusion [7] ships no implementation and [8] publishes code whose weights are no longer retrievable. Two are runnable and we report both: BiDAStabilizer [3], which refines a frozen estimator as we do but bidirectionally, and TC-Stereo [6], whose rigid pose warp needs a privileged input our interface does not assume but SCARED-C supplies, so we run it as its authors intended (Sec. V-I).

d) Evaluating temporal refinement.: Stereo benchmarks report frame-wise geometry and video methods add consistency measures, but neither characterises an intervention: a prediction can grow smoother without growing more correct, and a lower mean error can hide newly created failures. Recovered and introduced error must both be measured, which connects to selective prediction [16]. Our contribution is not a new metric but a common protocol under fixed, prediction-independent support.

e) Surgical stereo geometry.: SCARED [17] and DRENDS [21] supply the geometry we evaluate on; D4D [20] is only the substrate of Sec. VII's no-reference control, and SERV-CT [18], whose pairs are non-consecutive, is not used. Downstream reconstruction consumes exactly the per-frame geometry we refine [19], [25], which is why Sec. V-H measures after accumulation as well.

III. Method

A. External Causal Interface

A frozen estimator S produces positive-left disparity d_t^{raw} and its validity M_t^{raw}. The temporal module never reads the internal representation of S; it consumes only externally observable quantities — RGB, disparity, validity, optical flow and evidence derived from them. At time t the cache holds the preceding left image, the preceding raw disparity and validity, the previous fused state and its age; nothing from $t+1$ onwards exists. The map to learn is $d_t^{\mathrm{fused}} = T_\theta(d_t^{\mathrm{raw}}, m_{t-1}, I_t^L, I_t^R, I_{t-1}^L)$.

B. Causal Motion Alignment

Frozen SEA-RAFT [5], [22] estimates both directions between the two already observed left images; the reverse direction is used only to measure reliability, and bidirectional *flow estimation* between two observed frames is not bidirectional temporal inference. Previous geometry is pulled onto the current grid, $\widetilde{m}_t(x) = m_{t-1}(x + F_{t\to t-1}(x))$, by bilinear sampling with zero padding; warp support $\mathcal{S}_t$ records whether the source coordinate lies inside the preceding image, and historical validity is sampled on the same grid. Pulling the reverse flow the same way gives the forward–backward residual e_t^{FB} and the reliability cue $C_t^{FB} = \mathcal{S}_t \exp(-e_t^{FB}/\rho_t)$ with ρ_t scaled by the flow magnitudes. The zero padding is not a detail: it makes $\widetilde{m}_t$ exactly zero outside the source image, and Sec. V-K shows what happens when that value reaches the output.

C. Shared External Evidence

All estimators are mapped into one 38-channel tensor on the cache grid, built from raw disparity, aligned memory, the two images, the previous left image, the flow, C_t^{FB}, the warp support and the sampled historical validity. Twenty-five channels are disparity-geometry correlations formed at quarter resolution and 13 are formed on the grid directly (Fig. 1, right); normalisation is fixed before training (disparities clamped to $[0, 64]/64$, residual $\pm 16/16$, flow $\pm 32/32$, RGB to $[-1, 1]$). Nothing in this space is learned, no network evaluates it, and no ground truth, backbone identity or future frame enters it. An earlier configuration added 104 quarter-resolution channels derived from a frozen ResNet-18 [9] — 64 compressed feature maps and 40 confidence curves, supports and appearance correlations formed from them — for 142 in total. The channel ladder of Table III reads off this decomposition: dropping only the 64 feature maps gives 78, dropping all 104 is TETHER's 38, and dropping its 25 disparity correlations leaves the 13 direct channels. Sec. V-J reports the 142-channel configuration as an ablation, because it costs 157,504 frozen parameters and three forward passes per frame and does not survive a three-seed comparison. The full channel specification is in the extended version.

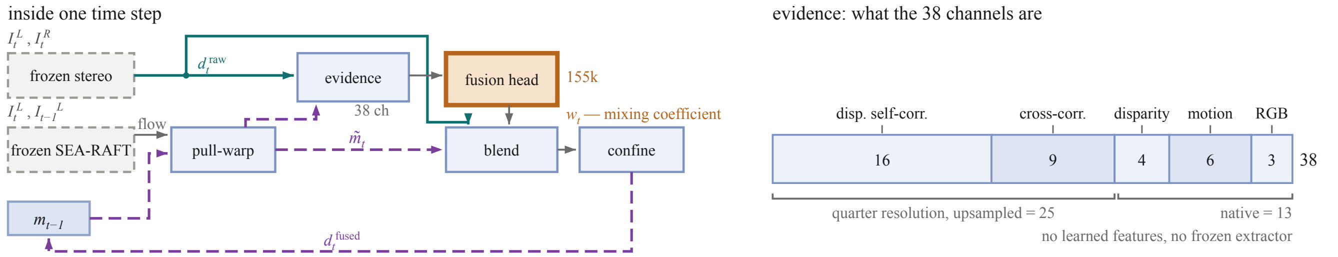

Fig. 1. One time step, and the evidence it reads. **Left:** dashed grey is frozen; the single orange block is the only trained component and emits one scalar per pixel, w_t, rather than geometry. **Right:** the 38 channels — 25 quarter-resolution disparity correlations, upsampled, and 13 native — none carrying ground truth, backbone identity or a future frame.

D. Reset-and-Fusion Decision Head

The two-gate decomposition of the temporal decision, and the constraint that the correction be convex in the two candidates, are CODD's [1]; the architecture, the motion pathway, the evidence space, the supervision and the training are not.

A cache-grid pathway and a quarter-resolution context pathway (Fig. 1) feed a full-resolution reset branch predicting $r_t = \sigma(z_t^r)$ and a coarse fusion branch predicting $f_t = \uparrow_4 [\sigma(z_t^f)]$, giving the effective temporal weight $w_t = r_t f_t$ from 154,874 trainable parameters. Running the context branch coarsely is a compute choice on that branch, $16\times$ cheaper than full resolution, which Sec. V-J reports as an ablation. Both branches run at width 48.

Temporal evidence is only defined where the aligned memory is usable, so the per-pixel *support* is the conjunction of current raw validity, aligned temporal validity and warp support,

$$S_t = M_t^{\text{raw}} \wedge \widetilde{M}_t \wedge \mathcal{S}_t, \tag{1}$$

and we confine the intervention to it. Final disparity is

$$\boxed{d_t^{\text{fused}} = \begin{cases} (1-w_t)d_t^{\text{raw}} + w_t \tilde{m}_t, & S_t = 1, \\ d_t^{\text{raw}}, & S_t = 0. \end{cases}} \tag{2}$$

Because $w_t \in [0,1]$ the fused disparity lies between the two candidates on the support, and off it the module is the exact identity on raw geometry. Applying (1) to the output and not only computing it is load-bearing (Sec. V-K). Reset output bias is initialized to -2 and fusion to zero, so initial intervention is conservative.

E. Bounded Corrected-State Recurrence

After refinement $m_t = d_t^{\text{fused}}$, so alignment and fusion errors can become future evidence. We bound corrected-state lifetime: the first output of a sequence is raw, a re-anchor restores previous raw disparity as state, and otherwise the previous fused output is propagated.

The shipped policy permits four corrected-state uses before a raw re-anchor, so H denotes state lifetime rather than a temporal input window; at lifetime H the mean age of the state in use is $(H+1)/2$ frames, the quantity Sec. V-H finds the accumulated cost tracks. The method remains one-step causal: $\{I_{t-1}, I_t, m_{t-1}\} \to d_t^{\text{fused}}$.

F. Training Objective

For target disparity g_t let $\Delta_t = |\tilde{m}_t - g_t| - |d_t^{\text{raw}} - g_t|$, so $\Delta_t < 0$ marks pixels where aligned memory is the better candidate; training support is the intersection of GT coverage, current validity, aligned temporal validity and warp support. The unweighted objective $\mathcal{L} = \mathcal{L}_{disp} + \mathcal{L}_{reset} + \mathcal{L}_{fusion}$ pairs a Huber term on the fused disparity with two decision terms supervising each branch against the sign of Δ_t at native margins 5 and 1 px, scaled to the cache grid, plus a weak pull of the fusion weight towards 0.5 where the two candidates are within the fusion margin. No explicit temporal-consistency objective is used: temporal behaviour is a consequence of the decision, never a training target.

IV. Experimental Protocol

A. Training

The temporal refiner is trained on temporally curated SCARED-C data derived from SCARED [17], using frozen predictions from S^2M^2-S [13], RAFT-Stereo [10] and Stereo Anywhere [12]; CREStereo [11] and Fast-FoundationStereo [14] are excluded from temporal-head training, and no backbone identity is ever given to TETHER.

Training uses D1, D3 and D6 with D2 for development and D7 held out, over 8,616 frames giving 8,606 causal pairs per backbone from three seen backbones, in non-overlapping $H = 4$ clips, batch size 4, 12 epochs of AdamW at 2×10^{-4} with weight decay 10^{-4} and gradient clipping 1.0, selecting on development fused EPE. The budget is not a truncation: selection lands at epoch 8, 11 and 11 across three seeds, all inside the schedule, and a locked-recipe run to 47 epochs oscillates without trend from epoch 13 with no later epoch improving on it. One ablation (A4) diverges after epoch 10, which best-validation selection catches.

B. External Evaluation

DRENDS [21] supplies dynamic robotic-endoscopy stereo with calibration and metric geometry; we report five curated recordings (7476 frames in total), with all five estimators. Its disparity reference comes from metric ToF geometry through stereo calibration, so it is useful metric evidence but not independent stereo ground truth, and the ToF reference is itself temporally filtered. The zero-shot claim applies to the SCARED-trained checkpoint: no DRENDS data enters training, selection or calibration.

TABLE I

ONE FROZEN TETHER CHECKPOINT, SEED 0, ON THE TWO ARENAS IT NEVER DEVELOPED ON; EVERY CELL IS RAW→TETHER, LOWER IS BETTER, AND THE LOWER-RIGHT BLOCK IS THE JOINT UNSEEN-BACKBONE, UNSEEN-DOMAIN SHIFT USUALLY LEFT UNMEASURED (* MARKS THE TWO ESTIMATORS EXCLUDED FROM TEMPORAL-HEAD TRAINING). DEVELOPMENT CELLS (SEC. V-A), DTCE COLUMNS (SEC. V-F) AND PER-SEQUENCE DETAIL (EXTENDED VERSION) COMPLETE THE FIFTEEN-CELL COUNT OF CONTRIBUTION (I). CELLS ARE POOLED PER BACKBONE.

	D7 held out		DRENDS, out of domain	
Backbone	EPE	Bad1 (%)	EPE	Bad1 (%)
S^{2}M^2-S	$.518 \rightarrow \mathbf{.499}$	$6.44 \rightarrow \mathbf{6.03}$	$1.348 \rightarrow \mathbf{1.281}$	$62.6 \rightarrow \mathbf{59.0}$
RAFT-Stereo	$.467 \rightarrow \mathbf{.430}$	$6.09 \rightarrow \mathbf{5.75}$	$1.313 \rightarrow \mathbf{1.237}$	$58.8 \rightarrow \mathbf{55.2}$
Stereo Anywhere	$.477 \rightarrow \mathbf{.441}$	$6.23 \rightarrow \mathbf{5.77}$	$1.369 \rightarrow \mathbf{1.283}$	$62.7 \rightarrow \mathbf{58.0}$
CREStereo*	$.517 \rightarrow \mathbf{.491}$	$6.35 \rightarrow \mathbf{6.05}$	$1.303 \rightarrow \mathbf{1.230}$	$55.2 \rightarrow \mathbf{52.2}$
Fast-FoundationStereo*	$.482 \rightarrow \mathbf{.465}$	$5.92 \rightarrow \mathbf{5.65}$	$1.284 \rightarrow \mathbf{1.223}$	$54.9 \rightarrow \mathbf{52.1}$

C. *Unified Temporal-Refinement Evaluation*

All temporal processing and evaluation run on a fixed 144×180 cache grid (SCARED-C is natively 1280×1024; median ground-truth disparity on the grid is 8.0 px), so disparities here are grid-scale pixels. Evaluation support cannot depend on the prediction: $M_{\text{eval}} = M_{\text{GT}} \cap M_{\text{protocol}}$, and invalid predictions on valid support are penalised rather than silently removed. Temporal fidelity is change error $\text{DTCE} = |\Delta\text{prediction} - \Delta\text{target}|$ on that support at lags $k \in \{1, 2, 4, 8\}$ (k=4 unless stated), scored against the *reference* change rather than stillness, so the degenerate replay of Sec. VII does not win it. Harm and benefit are the masses $H^+ = \mathbb{E}[\max(\delta e, 0)]$, $B^+ = \mathbb{E}[\max(-\delta e, 0)]$ of $\delta e = e^{\text{fused}} - e^{\text{raw}}$ with their rates; NewBad and RecoveredBad count pixels crossing a bad-pixel threshold either way; the family also carries percentile and mm-threshold geometry, per-frame tails and risk–coverage [16]. Prose reductions are macro — the mean of per-cell relative reductions — unless stated; pooled sums before the ratio; each table declares its own aggregation.

V. RESULTS

The claim is one module across backbones and domains, so the results are ordered that way: the transfer matrix, the seed evidence that it is stable, why it works, what it does to a map, and what it costs.

A. *Cross-Backbone Development Behaviour*

$H = 4$ improves all five estimators in mean D2 EPE — 1.41, 2.28, 8.19, 3.10 and 2.24% for S^{2}M^2-S, RAFT-Stereo, Stereo Anywhere, CREStereo and Fast-FoundationStereo — and all six unseen-backbone sequence groups improve; on D7 the mean excluding Stereo Anywhere is 5.15%. One D2 cell worsens, S^{2}M^2-S on keyframe 2, by 0.28%: the first of three sequence-level exceptions in this paper, the others being Vid13 (Sec. V-I) and the development map (Sec. V-H).

B. *Held-Out Cross-Backbone Confirmation*

We next evaluate the already frozen $H = 4$ policy on held-out D7. All five backbones improve, including both unseen during temporal-head training: mean EPE decreases $0.4921 \rightarrow 0.4651$ pooled (5.49%); per cell the reduction is 5.40%.

The improvement is not restricted to the mean: Bad1, RMSE and P99 decrease for every backbone on both splits, so the EPE gain is not bought by a worse bad-pixel rate or a broader tail, though new bad pixels do appear and are measured below. In absolute terms, held-out depth MAE is $2.79 \rightarrow 2.71$ mm (RMSE $5.33 \rightarrow 5.23$; Bad-2mm $32.5 \rightarrow 31.8$%). InvalidRate stays zero in all ten SCARED-C backbone–split evaluations — a property of SCARED-C, not of the module (Sec. V-K).

C. *External Zero-Shot Geometry on DRENDS*

We evaluate the frozen checkpoint on five DRENDS recordings with all five estimators, two excluded from temporal-head training, 7476 frames in total, nothing adapted or tuned for the domain. Disparity EPE, Bad3, metric depth MAE and RMSE improve in *every* one of the 25 recording–backbone cells, by -5.41, -19.51, -4.02 and -3.47% on average. Residual variance is set more by the recording than the estimator but not uniformly: the EPE change spans 0.87 points across backbones on Vid10 against 3.29 on Vid13. Absolute depth MAE is $14.7 \rightarrow 14.1$ mm: the out-of-domain regime is failed geometry improved, not good geometry polished.

Averaged over backbones the mean EPE reduction is 3.96% seen against 2.67% unseen on D2, 5.80 and 4.79% on D7, 5.57 and 5.16% on DRENDS: every combination positive. We can separate neither group from the other and report the margins rather than a test — with three seen and two unseen backbones and per-backbone margins from a single seed, absence of a detectable gap is not evidence of no gap. We read no ordering into the numbers either: on D2 the seen advantage is carried entirely by Stereo Anywhere and reverses without it, so we withdraw the claim we drew about which axis costs more. This establishes zero-shot geometric transfer under a ToF-derived reference; it does not establish universal robustness, and DRENDS remains a non-independent reference.

D. *Seed Variance, and What It Reveals About Selection*

Under a seed study pre-registered before any result existed (only the seed varies; selecting the best seed is prohibited), pooled over five backbones the relative Bad1 reduction is 12.73 ± 0.32% on D2 and 5.84 ± 0.22% on held-out D7; EPE gives 4.13 ± 0.38% and 5.61 ± 0.42%, where $\pm$ is the standard deviation over three seeds throughout this paper. The distribution-free statement the splits can carry is that the effect is positive in 3/3 and 4/4 sequences, a sign test at $p = 0.125$ and $p = 0.0625$. No split here reaches $p < 0.05$ and none can at these sequence counts: we claim consistency of direction, never significance.

The spread is the strongest argument for the shipped configuration, and not what we found first. The 142-channel ablation is stable held out (5.15 ± 0.34 Bad1) and unstable on development, its seeds falling monotonically over a 5.03-point range. We read that as selection optimism, since checkpoints are selected on D2. Removing the learned evidence removes the instability instead — 0.32 points of D2 seed deviation against 2.53, with no change to the selection rule — so the instability was the evidence space and not the protocol.

E. Does the Learned Temporal Policy Matter?

TABLE II

MATCHED-POLICY CLOSURE ON D2, SEED 0, FIFTEEN BACKBONE–SEQUENCE CELLS POOLED BEFORE EACH RATIO; REDUCTIONS IN %, HIGHER IS BETTER, DTCE AT k=4. EVERY ROW SHARES THE PREDICTIONS, ALIGNMENT, SUPPORT CONTRACT, H=4 STATE MACHINE AND EVALUATION CODE — ONLY THE POLICY CHANGES; THE FIRST SIX LOAD NO CHECKPOINT, AND UNBOUNDED RECURRENCE IS OUR OWN HEAD WITH THE RE-ANCHOR REMOVED. TETHER'S ROW IS SEED 0, THE WEAKEST OF ITS THREE (3.69 AGAINST 4.13 ± 0.38), SO EVERY MARGIN HERE UNDERSTATES IT.

Policy	EPE	Bad1	DTCE$_{k=4}$	NewBad1
Fixed w=0.5 ($\equiv$ EMA2)	0.59	−2.64	6.26	0.214
EMA, three frames	0.30	−8.88	4.88	0.379
FB-confidence as the weight	−2.17	−32.35	−7.10	0.831
Warped raw previous, H=1	−0.03	−6.39	−4.33	0.373
Tuned residual rule	0.87	0.33	2.88	0.088
Aligned-memory propagation	−3.56	−41.70	−13.55	1.072
Unbounded corrected recurrence	−19.17	−143.48	−28.13	2.858
TETHER, H=4	**3.69**	**12.84**	**4.99**	**0.063**
Raw-vs-memory oracle (ceiling)	12.74	21.87	10.01	0.000

We isolate the learned decision policy on D2, seed 0. All variants share the frozen predictions, the alignment, the support contract and the evaluation code, and all run unconfined so the comparison shares one implementation (confinement moves D2 EPE by 1.2×10^{-5} px). Figures are pooled over the fifteen backbone–sequence cells, whose raw EPE is 0.276202.

The fifteen policies (Table II) span the horizon sweep, fixed and EMA blends, confidence and propagation rules, and a tuned two-parameter residual rule swept over seventeen (α, τ) points. Per backbone, against whichever fixed or EMA policy is best *for that backbone*, the ratio ranges 3.9× to 9.9×: the absolute gain depends on the estimator, the margin over smoothing does not.

Three rows carry more than the mean. Forward–backward confidence is the informative failure — one of TETHER's own cues, yet used directly as the weight it worsens EPE by 2.17% and Bad1 by 32.35%. The uniform smoothing policies buy mean error with bad pixels, the more so the harder they blend: w=0.3 costs 0.18% more bad pixels for 0.49% of EPE, the best fixed blend 2.64% for 0.59%, EMA3 8.88% for 0.30%. The tuned residual rule limits that statement: a quarter of our gain, Bad1 improved. The trade belongs to blending *uniformly*: the moment a rule varies in space it stops buying mean error with tails, and the closest unlearned approach reaches 0.87% against our 3.69. What the learned head adds is not varying in space; it is knowing where. TETHER is the only row of Table II simultaneously first on mean EPE and lowest on NewBad1: every unlearned policy pays for its share of the mean in newly broken pixels — 3.4× ours for the best fixed blend, +40% even for the residual rule at a quarter of the gain.

The closure is development work, and a reviewer is right to ask whether the margin survives the split. We repeat it out of domain, on the arena the freeze permits: on DRENDS with RAFT-Stereo, the best unlearned policy recovers 0.63% of EPE against our 5.80% (RAFT-Stereo alone, pixel-pooled on the closure's support; the five-backbone per-cell mean is 5.41) — a ratio of 9× against 6× on D2. The denominator barely moves between arenas: uniform smoothing recovers the same small amount wherever it runs, so the widening is entirely ours, and it tracks how much raw error there is to recover (0.276 px on D2 against 1.286 out of domain, both on the closure's own support), the same quantity Sec. V-H finds decides the accumulated benefit. Where the raw prediction is this weak the uniform blends no longer pay in bad pixels (Bad1 +0.42, +0.53, +0.45% for fixed, EMA3 and FB-confidence), so that trade is stated on D2 alone.

F. Why $H = 4$, and Why Recurrence Must Be Bounded

Recurrence must be bounded: with no re-anchor the intervention inverts, EPE degrading by 19.17% and Bad1 by 143.48% (Table II). Within the bounded range EPE rejects H=1 by 1.20 points and H=2 by 0.67, and does not separate $H \in \{4, 6, 8\}$ (spread 0.21: 3.69, 3.73, 3.52%), while newly broken pixels rise monotonically from 0.052% to 0.128%. We froze H=4 before D7 opened on that risk–gain trade, and it is the last horizon at which change error also improves at lags $k \in \{2, 4, 8\}$ (4.99% at H=4 against −3.64% at H=6 for k=4). Two caveats. At k=1 DTCE improves at every horizon we tried, so the sign change belongs to the longer lags, not the metric. And on D2 the improvement is carried by one backbone and is *negative* for three: +14.0% for Stereo Anywhere and +0.8% for CREStereo against −0.3, −2.0 and −2.3%. A criterion whose sign one of five backbones decides cannot carry an operating point, so the horizon rests on breakage; on both confirmation splits DTCE is positive for every backbone (4.3–7.9% on D7, 8.5–13.9 on DRENDS). Sec. V-H sweeps the same horizon against accumulated geometry and finds the same risk structure from an independent measurement.

G. Refinement as an Intervention

Across both splits and all five backbones TETHER recovers 3.8–11.1× as many Bad1 pixels as it introduces, B^+/H^+ between 1.39 and 3.25. Per frame, 56% of the 15,535 held-out frames improve against 14% worsened (45 against 18% on development, the rest exact Bad1 ties on cleaner frames), the worsened band hugging the diagonal while the improved mass extends far from it: the count-versus-magnitude asymmetry behind B^+/H^+.

H. What a Map Fuses, Not What a Frame Scores

Everything above scores a frame. What a robot consumes is a map, so we measure that too: SCARED-C's corrected poses let geometry be accumulated in one world frame and the disagreement between views of a point measured, reported as a percentage of the excess over the ground-truth floor. On held-out D7 refinement reduces it by 11.7%, and the sign is not carried by one estimator: across five backbones and four sequences all 20 cells improve, by $+7.2$ to $+17.8\%$.

Copy-previous, which removes 97.3% of the no-reference score, is the *worst* policy of the sweep below: a replayed frame is wrong from the new viewpoint, so the measure survives the control contribution (iii) demands of temporal metrics.

The horizon sweep names the condition, on RAFT-Stereo across the complete grid (per-cell means over 3 D2 and 4 D7 cells, shared poses, accumulation and support): $H{=}1/2/4/6$ give $-3.6/-8.6/-15.5/-20.2\%$ on development against $+8.9/+10.4/+11.7/+12.2$ held out, while copy-previous posts $-54.0/+2.8$ and the ungated warp $-37.6/+7.2$. The five-backbone confirmation exists at the shipped horizon only (Sec. V-B), so the mechanism is established on one estimator and its sign confirmed on five. On D2 the cost grows monotonically with state lifetime and decelerates — 3.6% at $H{=}1$, one warp and no accumulation, to 20.2% at $H{=}6$ — so a map pays not for the warp but for carrying it. The ungated warp at full weight costs 37.6% where the gated head at $H{=}1$ costs 3.6%, and on D7 already returns 7.2 of our 11.7: on a map most of the gain is alignment and most of the gate's value is damage avoided, the same division Sec. VI finds per pixel. Since alignment residual is a property of the motion while raw excess differs twofold between the splits (0.171–0.290 mm against 0.518–1.113), whether that residual is bought back is decided by the split.

Both splits are monotone in H and neither turns, so the horizon is set by an exchange rate rather than a peak. Held-out benefit saturates while development cost keeps growing — successive steps buy $+1.5$, $+1.3$ and $+0.5$ points on D7 for -5.0, -6.9 and -4.7 on D2 — so the price of lifetime rises from 3.3 to 9.4 points of development error per point of held-out gain, making $H{=}6$, which the D7 column read alone would recommend, the worst purchase on the sweep. $H{=}4$ was frozen on the per-frame trade before this measurement existed and the map is the third criterion to place it there (frame EPE unreadable at 0.21 of spread, breakage rising, 4.7 paid per 0.5 gained). A consumer that only accumulates may still prefer shorter — $H{=}1$ returns 76% of the held-out benefit at under a quarter of the development cost, for 1.24 points of frame EPE — which makes the horizon a dial rather than an optimum.

I. Head-to-Head Against a Published Stabiliser

Against BiDAStabilizer [3] on 117.4M pixels of common DRENDS support — its own RAFT-Stereo *robust* array, a support distinct from Table I's — TETHER leads on every geometric metric (EPE $1.199 \rightarrow \mathbf{1.131}$ against 1.175; Bad1 $44.92 \rightarrow \mathbf{43.39}$ against 44.81) while being the only causal entry and 62× smaller in trained parameters. Two asymmetries run in opposite directions — the baseline is bidirectional, which our setting forbids, while we are trained on surgical endoscopy and it is not — so the margin is an upper bound on our advantage; the named exception is Vid13, where on this support our intervention is a net harm and the baseline's is not. The temporal measure reverses the ordering: the bidirectional baseline is markedly more stable at every lag and simultaneously less accurate on every geometric metric, this paper's own warning turned on its own comparison. TC-Stereo [6], run as its authors specify, is outside its domain before its temporal path runs (0.524 against 0.303 raw), and enabling that path makes both metrics worse (0.622, 10.37% Bad1).

J. Ablating Our Own Design Decisions

Four design choices were ablated under a pre-registration naming held-out Bad1 as the endpoint and a three-seed spread of 0.37 points as the threshold below which a difference is not read — the standard deviation over the base configuration's three seeds of its pooled Bad1 reduction across all 35 backbone–sequence cells, computed before the registration was written. Promotion to the shipped model is governed by a separate registration that decides on development only.

TABLE III

HELD-OUT RELATIVE REDUCTION (%), NOT ERROR: HIGHER IS BETTER, MACRO OVER BACKBONE–SEQUENCE CELLS. EACH INDENTED DEVIATION IS MEASURED AGAINST THE UN-INDENTED BASE ABOVE IT, AS ITS PRE-REGISTRATION NAMES. DROPPING THE LEARNED EVIDENCE FROM THE 142-CHANNEL BASE IS WHAT TETHER NOW IS, SO THE SHIPPED MODEL HEADS THE TABLE; ITS OWN FULL-RESOLUTION AND UNCONSTRAINED-OUTPUT DEVIATIONS ARE IN THE TEXT. EVERY ROW IS A THREE-SEED MEAN ± SD.

Configuration	Evidence	D7 EPE red.	D7 Bad1 red.
TETHER (3 seeds)	38 ch	$\mathbf{5.61 \pm 0.42}$	$\mathbf{5.84 \pm 0.22}$
geometry only (3 seeds)	13 ch	4.61 ± 0.21	5.36 ± 0.68
+ learned evidence (3 seeds)	142 ch	5.06 ± 0.44	5.15 ± 0.34
no appearance (3 seeds)	78 ch	5.10 ± 0.80	4.89 ± 0.31
full-res context (3 seeds)	142 ch	5.24 ± 0.52	5.68 ± 0.16
unconstrained output (3 seeds)	142 ch	1.70 ± 1.43	6.97 ± 0.83

The result (Table III) is not the one we expected. We began where the video-stereo literature does, with a frozen ResNet-18 [9] supplying appearance correlations and cost curves on top of the geometric channels — the configuration the four deviations were pre-registered against. The registered threshold assumes spreads comparable to the base's 0.37, and every deviation of the shipped head exceeds it (0.42–0.68), so we read rather than apply the rule there: none of the shipped head's three deviations separates. On the 142-channel base the rule does fire once — full-resolution context clears it against that base (below), declined on compute — while three deviations sit above their base (the shipped head among them) and dropping appearance sits inside; the shipped head's three — no FB cue (5.42 ± 0.42), geometry only (5.36 ± 0.68), full-resolution context (5.57 ± 0.68) — are unseparable from 5.84 ± 0.22, with seed 0 again at the low end, as it is for the shipped model on the closure and at the high end for the base — the pattern Sec. V-D names. On EPE the direction is consistent for every deviation and A6 comes closest to separating (4.61 ± 0.21 against 5.61 ± 0.42, under four standard errors): the evidence space shows on the mean where it does not show on the registered endpoint. Removing the learned evidence entirely, under the promotion rule that decides on D2 with D7 reported afterwards, stays ahead on both endpoints with seed deviation eight times tighter (0.32 against 2.53 points of D2 Bad1), and out of domain the ordering holds ($5.20 \pm 0.13\%$ against $4.83 \pm 0.15\%$ DRENDS EPE, three seeds each on three backbones). The learned evidence is not what makes the module work; it is what makes its training unstable, at 157,504 frozen parameters, three forward passes per frame and 13.8% of the runtime. Its apparent advantage on the development closure (4.05% against 3.69%) is a sharp seed-0 artefact: seed 0 is the *best* of three for it and the *worst* for us, so the single-seed pairing anyone would report is the worst available to us, and over three seeds the ordering reverses ($4.13 \pm 0.38\%$ against $3.55 \pm 0.45\%$).

Full-resolution context is the firing noted above: it wins the registered endpoint against its own base (5.68 ± 0.16 against 5.15 ± 0.34), is unseparable on the base we ship (5.57 ± 0.68), and is declined on compute.

Ablating convexity (A4) posts the largest held-out Bad1 reduction in the table, $6.97 \pm 0.83\%$, but its margin over us (1.13 points) is not separable from its own three-seed spread, four times ours. Its EPE loss *is* separable: $1.70 \pm 1.43\%$ against our 5.61 ± 0.42, over four standard errors. We decline it on the metric where the difference is real rather than on the metric that suits us. The two trajectories put side by side say what the constraint buys: from development to held-out, A4's EPE gain falls $5.32 \rightarrow 1.70 \pm 1.43$ while ours rises $4.13 \rightarrow 5.61 \pm 0.42$, and the drop holds in all fifteen backbone–seed pairs — the unconstrained output buys in-domain accuracy that does not survive the split shift, and its 38-channel twin repeats the pattern ($11.25 \rightarrow 4.41$ EPE, one seed). Its registered consequence we honour: the falsification clause required that if A4 matched or beat the canonical model held out, the claim attributing *cross-backbone* transfer to the convex action space be withdrawn rather than softened. It is withdrawn and appears nowhere in this paper; the cross-*split* statement above is a narrower claim, measured on A4's own three seeds.

K. The Support Contract Is Load-Bearing

Confining the output to (1), not merely computing it, is a single line. On DRENDS the difference is categorical: unmasked, the shipped head *regresses* depth RMSE in every one of the 25 recording–backbone cells, by $+221.9\%$ on average (range $+29.5$ to $+594.5$); confined, it improves in every one of the same 25 cells, by -3.5% (-6.7 to -1.8). Zero-padded out-of-bounds memory enters the blend as $\widetilde{m}_t = 0$ and bounded recurrence carries the collapse to the next re-anchor; the offenders are rare and bursty (300 of 38.8M pixels on one recording, aligned memory exactly 0.0 for every one) and, since invalid predictions on valid support are penalised rather than discarded, they are the entire source

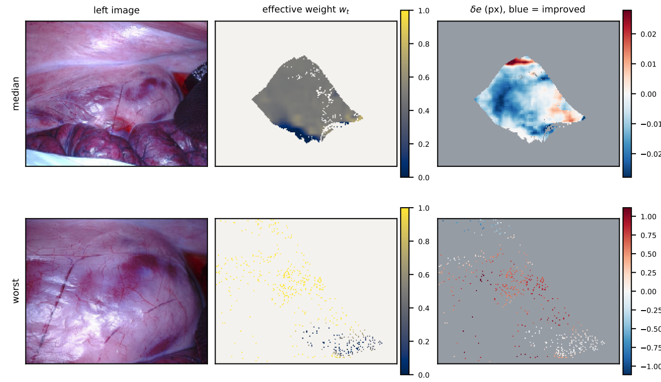

Fig. 2. Where the module intervenes, and whether it helped there. Grey and off-white are off support, where (2) makes the module the exact identity and δe is zero by construction, so those pixels are not drawn at the centre of the diverging scale. **Top:** the median frame of a D2 sequence — acts on 19.2% of pixels at mean weight 0.46, improves 82.8% of them. **Bottom:** the worst frame of the same sequence, by rank rather than choice — 2.2% at 0.79, improves 12.9% (the δe scales differ $40\times$ between rows). "Acts on" is the fraction of the frame's pixels on support. Few pixels, high weight, wrong is a tail and not a trend: over the sequence's 1,861 frames with defined weight, per-frame mean weight correlates *negatively* with harm (Spearman -0.68; -0.26, same sign, on a held-out sequence).

of a depth tail we had attributed to the external domain. On SCARED-C the same ablation is nearly inert: the contract costs a little in-domain gain, buys well-posed behaviour out of it, and is almost invisible to anyone validating in domain only.

VI. Analysis

Alignment is necessary but not sufficient: the aligned memory carries a 12.74% oracle ceiling that indiscriminate use does not collect (Table II). The mechanism is alignment plus learned graded intervention plus bounded recurrence, not smoothing; the head never solves stereo matching, it chooses between two candidates.

a) What each branch of $w_t = r_t f_t$ does.: Suppressing either factor at inference, the trained head unchanged, separates their jobs. Fusion alone keeps most of the gain, 3.49% against 3.69%, but breaks a third more pixels (0.084% New-Bad1 against 0.063%); reset alone keeps 2.23% and breaks 0.211%. The fusion weight sets how much is gained, the reset gate how little is broken, and a paper reporting only the product and only mean error would have seen neither. Sec. V-H reaches the same division from accumulated geometry. Ablating convexity improves RMSE and worst case while collapsing the mean EPE gain (Sec. V-J): the failure is broad degradation of pixels already close, so the constraint buys mean accuracy rather than safety from large corrections.

Two limits are worth stating. Against the raw-versus-memory oracle of Table II — measured on TETHER's own aligned memory, since under $H{=}4$ that memory is the previous fused output on three frames of four — TETHER recovers 29.0% of the available gain on D2 pooled and 25.2% per cell, so the open problem is deciding *where* to intervene.

And the effective weight w_t is an intervention magnitude rather than an uncertainty: read in the benign direction it ranks helped pixels at AUROC 0.59–0.61 on the split the checkpoint was selected on and at chance (0.49–0.51) held out — selection optimism (Sec. V-D) appearing a second time, with the same shape in the 142-channel configuration, so the limit belongs to the formulation. It does not contradict the intervention ratios, which need no per-pixel ranking: a policy can be good on average and uninformative per pixel.

VII. Discussion and Limitations

a) What the splits support.: One external domain does not establish universal robustness, and DRENDS supplies a temporally filtered ToF reference rather than independent stereo ground truth; the remaining scope caveats are stated where each result is.

b) What the metrics cannot say.: No-reference temporal consistency is gameable, shown on D4D [20], its only role here: the module reduces that score by 53.9%, copy-previous by 97.3%, warped-previous by construction to zero. The accumulated-geometry measure of Sec. V-H survives the same controls, which is why the robotics claim rests on it. Pull-warp support detects out-of-frame sampling but not internal disocclusion.

c) Where the intervention costs a map.: On development D2 the mapping measurement runs the other way: refinement *increases* world-frame disagreement by 15.5% at the shipped horizon, uniformly, all 15 cells negative. The account in Sec. V-H is consistent with every cell we have but is not proven: the two splits' excess ranges never overlap, so no backbone makes D2 as hard as D7, and within D2 the per-backbone ordering does not follow excess either. What the grid does establish is that the reversal is a property of the split and of the horizon, not of one estimator: a module refining a prediction already close to its reference can cost a map more than it returns.

d) Cost, and where it is.: The head is compact but the deployed system also runs frozen stereo and two SEA-RAFT passes: 28.90 ms overhead at 95.2 MB peak VRAM on an H100. The frozen estimator dominates, so the end-to-end rate is set by the backbone, 11.6 FPS on Fast-FoundationStereo (overhead 33.5%) to 1.4 on StereoAnywhere (4.0%); *no real-time claim is made*. Stage timing locates it: the two SEA-RAFT passes are 26.2 of the 28.9 ms, against 0.67 for the evidence tensor and 1.00 for the head; the tensor is that cheap because TETHER builds no learned features, where the 142-channel ablation spends 5.13 ms there. The reverse pass exists only to compute C_t^{FB}, and the trained head barely reads it: pinned to a constant at inference, the within-checkpoint held-out delta over all three shipped seeds is $+0.02 \pm 0.06$ Bad1 and -0.09 ± 0.06 EPE, and a head trained *without* the cue is equally indifferent to receiving the real one (-0.02; one checkpoint), so the indifference is architectural. Training without it trends lower without separating at three seeds (5.42 ± 0.42 against 5.84 ± 0.22): **train with it, deploy without the reverse pass**, taking the module from 28.9 to ~ 15.8 ms on the unchanged checkpoint. We evaluate perception geometry and what a map accumulates from it, not downstream control, and demonstrate neither clinical safety nor patient benefit.

VIII. Conclusion

TETHER attaches one causal temporal module, trained once, to frozen and heterogeneous stereo estimators without touching their internals or using future frames. Its gain is not temporal averaging: matched fixed and EMA blends under an identical state machine recover a sixth of it on development — paying in bad pixels there — and a ninth out of domain. One frozen $H = 4$ checkpoint improves all five estimators on held-out D7 and all 25 recording–backbone cells zero-shot on DRENDS.

Five results bound what may be claimed, and they are why the protocol is part of the contribution: no-reference consistency is won by replaying warped history while the accumulated-geometry measure is not; the fusion weight ranks harm only on the split it was selected on; selection optimism belongs to the evidence space, not the protocol; a module that declares a support must apply it to what it emits; and what a map fuses improves held out while degrading on development, the candidate term being accumulated alignment residual — the horizon an exchange rate, not an optimum. What remains open is *where* to intervene. An extended version with per-sequence tables, the full policy grid, protocols and the complete channel specification accompanies the code, checkpoint and SCARED-C curation release.

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